
Out of the Loop: Bounded LLM Mediation for Cyber-Physical Control

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Abstract

Integrating large language models (LLMs) into the control loops of cyber-physical systems (CPS) exposes a structural tension: LLMs offer semantic reasoning over heterogeneous operational regimes, but their hallucinations propagate freely when placed inside the loop, whether as direct controllers, decision overrides, reward generators, or per-agent reasoners. Existing mitigations reduce hallucination probability but cannot bound its worst-case consequences, an architectural limit no probabilistic technique can resolve. We propose a different position: LLMs as bounded mediators alongside the control loop, elevated to system-level coordination reasoning that scalar rewards and per-agent reasoners cannot reach, while structural channels insulate the controller from LLM error. We instantiate this position as DAMA (Dual-Agent Mediation for Adaptive incentive design): a Sensing Agent interprets macro-state regimes and a Control Agent issues a coordination schedule of exclusion, priority, and negotiation sets, influencing any base controller only through action masking and reward shaping. We prove that per-step reward perturbation and the masked-agent fraction are bounded by framework hyperparameters regardless of LLM output, lifting to a bounded value loss against the base controller. Experiments across six demand-penetration regimes with three heterogeneous controller backbones show that DAMA matches or exceeds representatives of three LLM-in-loop patterns under nominal LLMs, uniquely holds the base-controller floor under five adversarial- and five hallucinated-LLM settings, and transfers zero-shot across unseen geometries and a multi-robot domain, confirming that the architectural interface is what carries.

1 Introduction

A growing body of work integrates large language models (LLMs) into the control loops of cyber-physical systems (CPS), which span domains from traffic and robotics to power grids and networked infrastructure, with the goal of leveraging their semantic reasoning over complex operational contexts. Yet across domains, LLMs are consistently placed where their correctness becomes a precondition for system safety: as direct controllers issuing actions, as decision overrides revising reinforcement learning outputs, as reward generators shaping agent behavior, or as per-agent reasoners negotiating maneuvers among themselves. In every case, an LLM hallucination propagates into the control loop: the very risk that has limited the deployment of LLM-based control in safety-critical CPS.

A second, parallel limitation arises from how the role of LLMs is scoped within these systems. Existing LLM-based control frameworks predominantly target agent-level objectives, such as safe maneuvers, local conflict resolution, individual trajectory quality, while system-level performance, such as throughput, capacity utilization, and congestion mitigation at bottlenecks in traffic, or coordinated load balancing in robotic fleets and energy systems, remains the domain of classical methods without LLM involvement. This allocation under-utilizes LLMs' comparative advantage: agent-level control is a kinematic, reactive problem with mature classical and learned solutions, whereas system-level coordination demands regime recognition, qualitative pattern interpretation,

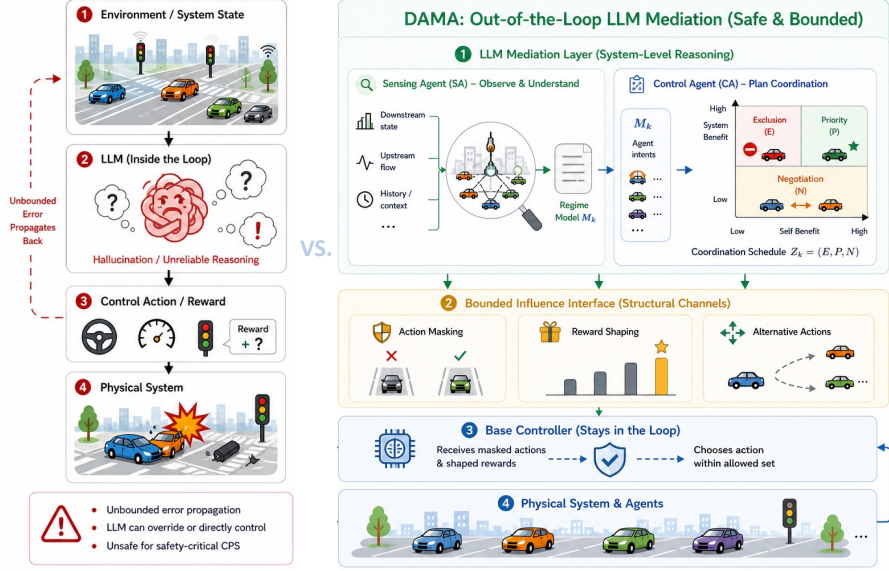


Figure 1: **LLM-in-the-loop vs. DAMA’s out-of-the-loop mediation.** *Left:* placing the LLM inside the loop lets hallucinations propagate unboundedly into control actions. *Right:* DAMA positions two LLM agents: a Sensing Agent (SA) emitting a regime model M_k and a Control Agent (CA) emitting a coordination schedule alongside the loop, influencing the base controller only through bounded channels: action masking, reward shaping, and alternative-action solicitation.

and contextual reasoning across heterogeneous signals, precisely where LLM capabilities are most distinctive relative to scalar-reward formulations.

We argue these limitations share a common architectural cause: LLMs have been placed inside the control loop, where they must be reliable for the system to be reliable, and where their high-level reasoning is consumed by low-level decisions. We propose the opposite position as a general design pattern for LLM integration in CPS: LLM operates as a mediation layer alongside the underlying controller, influencing system behavior only through structurally bounded channels and never substituting for the controller’s own decisions. Because the channels are bounded by construction, the influence of any single LLM call is bounded by framework hyperparameters rather than LLM quality, yielding graceful degradation under LLM error and freeing the LLM to focus on system-level coordination that scalar rewards cannot express. We instantiate this pattern as **DAMA (Dual-Agent Mediation for Adaptive incentive design)** and evaluate it primarily on mixed-autonomy traffic at bottleneck regions, with additional zero-shot transfer to a multi-robot warehouse task to probe the generality of the bounded-channel interface. Our main contributions are:

- **Repositioning LLMs in CPS control.** Rather than placing the LLM inside the loop where hallucinations propagate into safety-critical decisions, we position it as a bounded mediator alongside the loop, contributing semantic reasoning without decision authority.
- **Dual-agent mediation architecture.** We instantiate this as DAMA: two LLM agents emit a coordination schedule reaching the base controller only through action masking and magnitude-bounded reward shaping, leaving safety and feasibility with the controller and freeing the LLM for system-level coordination beyond scalar rewards.
- **Hyperparameter-bounded worst-case guarantee.** We prove that performance loss under arbitrary LLM output is bounded by framework hyperparameters alone, which is independent of LLM correctness, prompt, or model, and complementary to probabilistic mitigations such as chain-of-thought or output verification.
- **Evidence that the interface, not the domain, transfers.** Across six scenarios and three base controllers, DAMA matches or exceeds all four LLM-in-loop patterns nominally, uniquely holds the base-controller floor under adversarial and hallucination settings, and transfers zero-shot to two unseen geometries and a multi-robot domain.

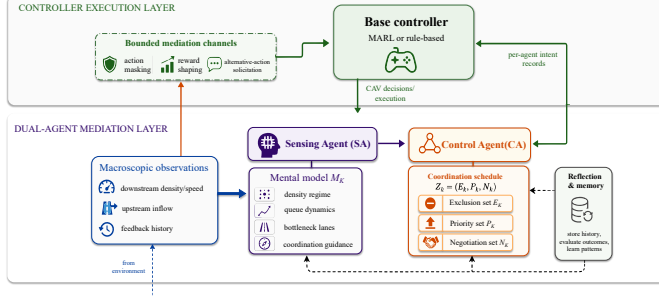


Figure 2: **DAMA architecture.** SA interprets macroscopic observations into a mental model M_k . CA combines M_k with per-agent intent records to emit a coordination schedule Z_k . Reflection and memory modules record each interval’s outcome and supplies retrieved episodes to both agents. The base controller receives Z_k only through three bounded channels.

2 Related Work

2.1 LLMs Inside the Control Loop and the Unresolved Hallucination Problem

A growing body of work integrates LLMs into control loops across cyber-physical systems, most actively in traffic and autonomous driving. Existing approaches differ in where the LLM sits, but share the structural property that LLM outputs influence control decisions without an architectural barrier between LLM error and system action. We group them into four patterns. (i) *LLM as direct controller*: the LLM issues control actions such as signal phases, lane changes, or trajectories [1, 2, 3, 4, 5, 6, 7]. (ii) *LLM as decision overrider*: the LLM inspects and revises an RL or classical controller’s initial decision when judged unreasonable [8, 9, 10]. (iii) *LLM as reward generator or shaper*: the LLM produces reward functions or shaping signals for RL training [11, 12, 13, 14]. (iv) *LLM as per-agent reasoner*: each controlled entity hosts an LLM that perceives, reasons, and negotiates with neighbors [15, 16, 17, 18].

These works adopt a range of mitigations against hallucination, including chain-of-thought [15], output verification [18], fine-tuning [1], and retrieval-augmented memory [15], driving residual rates as low as around 1% [18]. These share an epistemic limit: they reduce hallucination probability without eliminating it, and even chain-of-thought reasoning can be unfaithful to the model’s actual computation [19]. In safety-critical control, the relevant question is not whether the expected rate is small but whether the worst case is bounded. A 1% residual applied to millions of daily decisions yields thousands of erroneous outputs, any of which may coincide with a safety-critical moment. We argue the gap is architectural: as long as LLM outputs flow into control decisions without bounded channels, no prompt engineering or fine-tuning provides worst-case guarantees.

2.2 LLMs for System-Level Coordination

A smaller body of work uses LLMs at the system level, addressing coordination across multiple controlled entities. Network-level traffic signal coordination [2, 20], intersection-level conflict orchestration [3], fleet-level routing and scheduling suggestions [21, 22], and real-time traffic flow optimization through LLM-RL hybrids [23] all aim at system performance metrics such as throughput, congestion mitigation, or load balancing. These works confirm that LLM semantic reasoning is useful at the coordination layer; however, they retain the structural issue of the previous section, where LLM output (signal phase, intersection ordering, route assignment) is the action, so LLM error remains free to propagate. Closing this gap requires an architectural mechanism that bounds the worst-case effect of LLM output on system behavior while preserving the controller’s safety authority.

3 DAMA Framework

The architecture applies to a class of cyber-physical coordination problems we call upstream-controlled bottleneck coordination, which abstracts the structural features common to bottleneck-induced congestion in traffic, fleet routing through capacity-constrained corridors, and load-balanced task allocation in robotic and energy systems. The system is partitioned into a downstream evaluation region D , where the system-level performance metric is realized and capacity is finite and locally inelastic, and an upstream decision region U , where controllable agents operate with non-trivial action space before reaching D . The asymmetry is causal: agent decisions in U propagate into D and shape the metric there, but D ’s state is not directly actionable. Coordination reduces to: given

the current state of D , which agent actions in U should be encouraged, prevented, or left to the underlying controller, in order to optimize D 's performance?

DAMA is a mediation layer placed between the environment and a controller-agnostic execution layer. It consists of two LLM agents: a Sensing Agent (SA) and a Control Agent (CA), and each organized internally as a memory, a reasoning, and a reflection module. Figure 1 traces the data flow over one interval. The controller-agnostic execution layer is any policy mapping an agent's observation and admissible action set to an action distribution—a learned MARL policy, a rule-based controller, or an optimization-based scheduler.

3.1 Sensing Agent

Inspired by [24], we develop SA as a shared mental model $\mathcal{M}_k$ that the rest of the system coordinates against. Formally, SA is a mapping

$$\text{SA} : (D_k, U_k, F_k, \mathcal{H}_{k-1}) \mapsto \mathcal{M}_k,$$

where D_k is the downstream-region observation at interval k (per-lane density and speed trajectories over a short window), U_k is the upstream-region observation (per-lane net inflow), F_k is the aggregated feedback record from the previous interval (cooperation rate, realized downstream speed change, and CA's prior schedule statistics), and $\mathcal{H}_{k-1}$ is the previous memory and reflection. The output $\mathcal{M}_k$ is a structured semantic object carrying three classes of information needed for agent coordination: (i) *physical structure*: the salient regime descriptors of D ; (ii) *collective guidance*: the team-level coordination goal and the protocol under which CA should evaluate agent actions; and (iii) *linguistic interpretation*: a free-text rationale citing the observations behind the structural choices, used for audit and feedback consistency. Concrete fields for the traffic instantiation (regime axes, bottleneck lanes, prevention weight, risk warnings) are listed in the Appendix 8.3.

3.2 Control Agent

CA converts SA's guidance and the agents' local intents into the coordination schedule,

$$\text{CA} : (\mathcal{M}_k, \{\chi_i^k\}_{i \in I_k^U}, \mathcal{H}_{k-1}) \mapsto Z_k = (E_k, P_k, N_k),$$

where each χ_i^k is a side-output of Π summarizing agent i 's physical state, local context, and intended action with self-assessment (confidence, urgency, risk). Z_k partitions a subset of I_k^U into three pairwise disjoint sets; agents not assigned to any set receive the base reward and admissible action set unchanged. Schemas for χ_i^k and Z_k are in Appendix 8.4.

For each active agent, CA crosses two quantities—the *system value* of the intended action under $\mathcal{M}_k$ and the agent's *self value* from χ_i^k —conditioned on retrieved experience $\mathcal{H}_{k-1}$. The four resulting quadrants map to distinct mediations: both high $\rightarrow P_k$ (priority, action reinforced); system high, self low $\rightarrow N_k$ (negotiation, feasible substitute solicited); system negative, self high $\rightarrow E_k$ (exclusion, action vetoed); both low $\rightarrow$ no-op. This *agreement gate* realizes the principle of matching system intent against individual willingness; the channel structure below does not depend on the specific reasoning mechanism.

Z_k acts through three structurally bounded channels on Π . *Action masking*: for $i \in E_k$, the harmful action is removed from Π 's admissible set. *Reward shaping*: for $i \in P_k$ executing the prioritized action, a bonus $+\eta_P$ is added to the per-step reward. *Alternative-action solicitation*: for $i \in N_k$, a smaller bonus $+\eta_N \leq \eta_P$ applies to any action in a designated cooperative subset, letting Π substitute a feasible alternative when one exists. The hyperparameters $(\eta_P, \eta_N, \kappa_E)$ cap the per-step reward bonus and the masked-agent fraction.

3.3 Reflection and Memory Module

Following the reflect-then-memorize pattern of agent literature [25, 26], DAMA invokes a reflection LLM call after each interval's controller execution that diagnoses the prior joint decision against the realized outcome:

$$(\mathcal{M}_{k-1}, Z_{k-1}, F_k) \mapsto (\rho_{k-1}^{\text{SA}}, \rho_{k-1}^{\text{CA}}),$$

where SA's prior mental model $\mathcal{M}_{k-1}$, CA's prior schedule Z_{k-1} , and the realized outcome F_k (cooperation rate, downstream speed change, intervention statistics) together describe what was

155 decided and what happened. The module produces a structured diagnosis for each agent: ρ_{k-1}^{SA}
 156 assessing whether SA’s regime interpretation was supported by outcome, and ρ_{k-1}^{CA} assessing whether
 157 CA’s schedule was appropriately calibrated. The full triple $(\mathcal{M}_{k-1}, Z_{k-1}, F_k, \rho_{k-1}^{\text{SA}}, \rho_{k-1}^{\text{CA}})$ is written
 158 to memory as the episode e_{k-1} , where it becomes available for retrieval in subsequent intervals.
 159 Furthermore, DAMA maintains a shared memory module $\mathcal{H}_k$ that records the system’s coordination
 160 history and provides structured retrieval to SA and CA. The module exposes two operations: an update
 161 that appends a new episode after each reflection $(\mathcal{H}_{k-1}, e_{k-1}) \mapsto \mathcal{H}_k$, and two retrieval policies
 162 that surface relevant prior episodes for each agent at the start of the next interval $\mathcal{H}_{k-1} \mapsto \tilde{\mathcal{H}}_k$.

163 3.4 DAMA Control Loop

164 Algorithm 1 gives the full DAMA control loop. SA and CA are invoked asynchronously with per-call
 165 timeouts. On timeout or schema violation, the corresponding fallback applies, where SA falls back to
 166 a neutral mental model, and CA falls back to the empty schedule. Either fallback reduces DAMA to
 167 Π with its base reward, preserving baseline behavior under arbitrary LLM failure modes.

Algorithm 1 DAMA control loop. On any LLM timeout or schema violation, the corresponding module returns its neutral output $(\mathcal{M}^\emptyset, Z^\emptyset, \text{or neutral } \rho)$ and the loop continues unchanged.

Require: SA, CA, Reflect, memory $\mathcal{H}$, base controller Π , interval Δt_c

```

1 Initialize  $\mathcal{M}_0 \leftarrow \mathcal{M}^\emptyset, Z_0 \leftarrow Z^\emptyset, F_0 \leftarrow \emptyset, \mathcal{H}_0 \leftarrow \emptyset$ 
2 for  $k = 1, 2, \dots$  do
3   if  $k \geq 2$  then ▷ reflect on prior interval, write to memory
4      $(\rho_{k-1}^{\text{SA}}, \rho_{k-1}^{\text{CA}}) \leftarrow \text{Reflect}(\mathcal{M}_{k-1}, Z_{k-1}, F_k)$ 
5      $\mathcal{H}_{k-1} \leftarrow \text{Update}(\mathcal{H}_{k-2}, (\mathcal{M}_{k-1}, Z_{k-1}, F_k, \rho_{k-1}^{\text{SA}}, \rho_{k-1}^{\text{CA}}))$ 
6   end if
7    $\tilde{\mathcal{H}}_k^{\text{SA}}, \tilde{\mathcal{H}}_k^{\text{CA}} \leftarrow \text{Retrieve}(\mathcal{H}_{k-1})$  ▷ retrieve relevant episodes
8    $\mathcal{M}_k \leftarrow \text{SA}(D_k, U_k, F_k, \tilde{\mathcal{H}}_k^{\text{SA}})$  ▷ mental model
9    $Z_k \leftarrow \text{CA}(\mathcal{M}_k, \{\chi_i^k\}_{i \in \mathcal{I}_k^U}, \tilde{\mathcal{H}}_k^{\text{CA}})$  ▷  $\chi_i^k$  from  $\Pi$ ; coord. schedule
10  Apply  $Z_k$  to  $\Pi$  via masking ( $E_k$ ), shaping ( $P_k$ ), solicitation ( $N_k$ )
11  Execute  $\Pi$  for  $\Delta t_c$ ; aggregate outcomes into  $F_{k+1}$ 
12 end for

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168 4 Theoretical Properties

169 We model the underlying control problem as a Markov game $\mathcal{M} = (\mathcal{S}, \{\mathcal{A}_i\}_{i \in \mathcal{I}}, P, R_{\text{base}}, \gamma)$, with
 170 $\mathcal{A}_i(s) \subseteq \mathcal{A}_i$ the admissible action set at s as determined by Π and the environment. Let R_{DAMA}
 171 and $\mathcal{A}^{\text{DAMA}}$ denote the modified reward and admissible action map under DAMA, and let π_{base}^*
 172 and π_{DAMA}^* denote optimal policies under $(R_{\text{base}}, \mathcal{A})$ and $(R_{\text{DAMA}}, \mathcal{A}^{\text{DAMA}})$ respectively, holding the
 173 LLM-produced schedule sequence $\{Z_k\}$ fixed. Write $V_R^\pi(s) = \mathbb{E}_\pi[\sum_{t=0}^\infty \gamma^t R(s_t, a_t) \mid s_0 = s]$.

174 **Lemma 1** (Per-step bounded influence). *For any state s , joint action a , and any LLM-produced*
 175 *schedule Z_k satisfying the validation rules,*

$$|R_{\text{DAMA}}(s, a) - R_{\text{base}}(s, a)| \leq \eta_P + \eta_N \text{ per agent}, \quad \frac{|\{i \in \mathcal{I}_k^U : \mathcal{A}_i^{\text{DAMA}}(s) \neq \mathcal{A}_i(s)\}|}{|\mathcal{I}_k^U|} \leq \kappa_E.$$

176 **Theorem 1** (Bounded worst-case value loss). *For any state s and any LLM-produced schedule*
 177 *sequence $\{Z_k\}$ satisfying the validation rules,*

$$|V_{R_{\text{base}}}^{\pi_{\text{DAMA}}^*}(s) - V_{R_{\text{base}}}^{\pi_{\text{base}}^*}(s)| \leq \underbrace{\frac{\eta_P + \eta_N}{1 - \gamma}}_{\text{shaping slack}} + \underbrace{\kappa_E V_{\text{max}}}_{\text{masking slack}} =: \varepsilon(\eta_P, \eta_N, \kappa_E),$$

178 where $V_{\text{max}} = R_{\text{max}}/(1 - \gamma)$.

179 The Lemma 1 follows directly from the channel definitions; the theorem decomposes the bound
 180 into shaping and masking contributions and combines them via the triangle inequality. Full proofs
 181 are in Appendix 8.5 and 8.6. The bound $\varepsilon(\eta_P, \eta_N, \kappa_E)$ is a ceiling the designer sets in advance,
 182 independent of the LLM’s correctness, prompt, or model choice, which is a guarantee orthogonal to
 183 probabilistic mitigations such as chain-of-thought or output verification.

Table 1: In-distribution scenarios on the weaving-ramp geometry.

Scenario	Description	Demand (veh/h)	CAV penetration
I	Low demand	7,200	50%
II	Low penetration	12,000	30%
III	Medium	12,000	50%
IV	High penetration	12,000	80%
V	High demand	15,600	50%
VI	Peak demand	18,000	50%

Table 2: In-distribution comparison on Scenario III ($n = 10$ seeds, mean $\pm$ std). Best in **bold**, second-best underlined.

Method	Throughput $\uparrow$	Collisions $\downarrow$	Em. brakes $\downarrow$	TTC P5 (s) $\uparrow$	TTC min (s) $\uparrow$
<i>Non-LLM baseline</i>					
NLC	20.4 ± 1.4	0	130 ± 21	0.128	0.064
<i>LLM-in-loop framework baselines</i>					
LLM-Direct [1]	16.9 ± 1.8	38	182 ± 24	0.082	0.033
LLM-Override [8]	15.8 ± 1.7	2	209 ± 15	0.051	0.016
LLM-Reward [11]	<u>19.7 ± 1.5</u>	0	181 ± 33	0.127	0.065
<i>Our framework</i>					
DAMA	21.2 ± 1.4	0	132 ± 20	<u>0.125</u>	<u>0.056</u>

5 Experiments

We evaluate four claims: (i) bounded mediation matches the non-LLM upper envelope while LLM-in-loop patterns trade flow for safety; (ii) Theorem 1’s bound holds under adversarial and hallucinated LLMs; (iii) the bounded-channel interface transfers across base controllers, geometries, and domains; (iv) SA and CA contribute independently and super-additively. Six weaving-ramp scenarios in SUMO (Table 1); 80 s episodes. SA and CA use GPT-4o (JSON, $\Delta t_c=5$ s, 2 s timeout). Default hyperparameters $\eta_P=0.10$, $\eta_N=0.05$, $\kappa_E=0.20$. Three LLM-in-loop representatives: LLM-Direct [1], LLM-Override [8], LLM-Reward [11], and the structural lower bound NLC. A minimal codebase reproducing Table 2 is included as supplementary material (dama_minimal.zip).

5.1 Performance against Three LLM-in-Loop Patterns

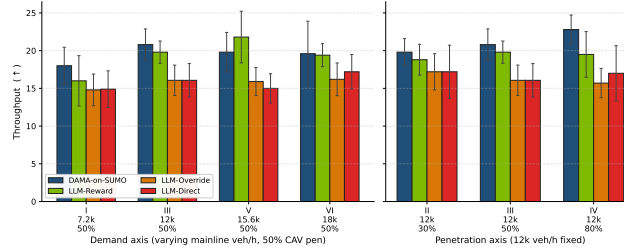


Figure 3: Throughput across six in-distribution scenarios.

DAMA is the only method attaining top-tier throughput and top-tier safety simultaneously (Table 2): flow matches the non-LLM upper envelope and emergency-braking is within 1% of NLC. The three LLM-in-loop frameworks order with their structural constraint on LLM output: LLM-Direct $\preceq$ LLM-Override $\preceq$ LLM-Reward $\preceq$ DAMA. The hierarchy reproduces across all six demands (Figure 3); the gap to the next-best LLM-in-loop framework widens at high demand, where coordination leverage dominates. Furthermore, DAMA performs better under higher CAV penetration rates.

5.2 Robustness under Adversarial LLM

The same degradation is applied to all four LLM-using frameworks, isolating architecture from model quality. We replace the LLM with five adversarial sources: **A1** random; **A2** anti-aligned (cost-maximising, schema-valid); **A3** max-intervention (saturate the framework’s budget); **A4** malformed (50% schema violations); **A5** timeout. DAMA’s worst-case throughput deviation stays an order of

Table 3: Throughput under five adversarial-LLM settings on Scenario III ($n = 10$ seeds).

Setting	DAMA	LLM-Direct	LLM-Override	LLM-Reward
<i>GPT-4o reference</i>	19.60 ± 1.51	16.50 ± 2.42	16.40 ± 2.32	19.70 ± 1.49
A1 (random)	19.90 ± 1.73	17.50 ± 2.37	16.90 ± 2.47	19.80 ± 1.55
A2 (anti-aligned)	20.60 ± 1.78	17.70 ± 2.83	18.90 ± 2.51	19.70 ± 1.64
A3 (max-intervention)	19.10 ± 2.47	20.40 ± 2.22	19.80 ± 1.87	19.30 ± 1.57
A4 (malformed)	19.30 ± 1.49	17.20 ± 2.44	16.70 ± 1.64	19.70 ± 1.49
A5 (timeout)	19.60 ± 1.51	17.60 ± 2.72	16.00 ± 2.26	19.60 ± 1.51
<i>Stability summary</i>				
$\max \Delta $ vs nominal	1.00	3.90	3.40	0.40
range across A modes	1.50	3.20	3.80	0.50

205 magnitude below the unbounded LLM-in-loop wrappers (Figure 4); LLM-Reward, the only LLM-
 206 in-loop pattern with magnitude bound, is comparable on this metric, but its architectural distinction
 207 surfaces under hallucination.

208 Sweeping η_P , κ_E , and the SUMO-LC realisation `lcSpeedGain` under A3 (Table 15) confirms that
 209 the bound of Theorem 1 is sound across all settings and tight at the defaults: throughput is invariant
 210 DAMA-internally, and $\eta_P=0.20$ introduces collisions absent at $\eta_P=0.10$. This single transition
 211 identifies the default as a safety frontier rather than an arbitrary choice.

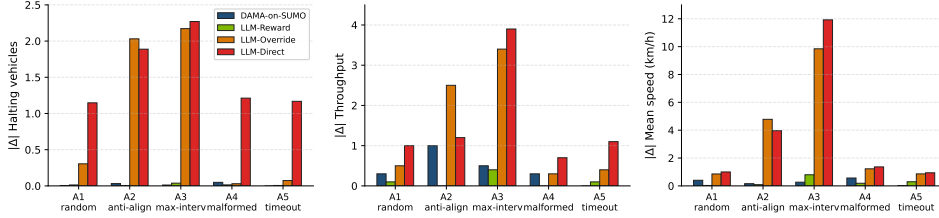


Figure 4: Performance across five adversarial settings.

212 5.3 Robustness under LLM Hallucination

213 The danger of hallucination that the LLM output passes schema validation: it is a plausible macro-
 214 state that does not match the underlying traffic. Schema validators cannot screen it, making this the
 215 most operationally relevant deployment failure mode. We construct five drivers, each corrupting
 216 SA’s output at every CA tick: **H1** “all clear” (under-report); **H2** “phantom jam” (over-report); **H3**
 217 mis-target (per-lane roles rotated by one index); **H4** drift (ζ_2 shifted $+0.30$ per tick); **H5** stale-context
 218 (first-tick output replayed forever).

219 DAMA’s halting deviation is exactly 0.00 across all five drivers (Table 5): CA’s regime validation
 220 rejects schema-valid but factually-wrong SA outputs and falls back to the empty schedule. The
 221 unbounded LLM-in-loop wrappers deviate one to two orders of magnitude more on every metric.
 222 LLM-Reward’s deviations are uniformly small but non-zero, which is because bounded magnitude
 223 limits the residual, but without a gate the bias persists every tick.

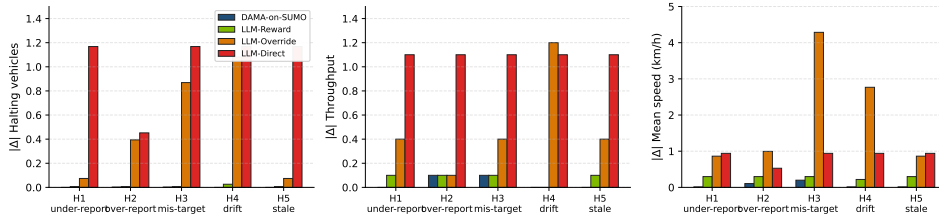


Figure 5: Performance across five hallucination settings.

224 5.4 Cross Controller, Geometry and Domain Transfer

225 Bounded mediation requires only an action-mask interface (E_k) and a parameter or reward-modulation
 226 interface (P_k, N_k). This subsection tests whether that decoupling holds across structurally different
 227 base controllers, unseen geometries, and other CPS domain.

Cross-controller transfer. We show safety preservation of DAMA under three base controllers: a rule-based gap-acceptance controller, the SUMO LC2013 cooperative model, and a MAPPO controller (MAPPO training setting is shown in Appendix 8.7), and measure how much hallucination perturbs each. Across all three backbones, DAMA’s hallucination-induced $|\Delta|$ stays below $\sigma/3$, where σ is the natural seed-to-seed variation of the same backbone (Table 4). The H3 perturbation is statistically indistinguishable from random-seed noise, which is the empirical signature of the framework-level bound.

Table 4: DAMA’s safety preservation under hallucination (H3, Scenario III, $n=10$ seeds).

Base controller	$ \Delta $	Seed noise σ	$ \Delta /\sigma$
Rule	3.1	32.3	0.10
SUMO LC2013	7.4	23.7	0.31
MAPPO	1.3	27.7	0.05

Table 5: Hallucination robustness on Scenario III ($n=10$ seeds, 80 s episode).

Metric	max $ \Delta $ from each framework’s own nominal			
	DAMA	LLM-Direct	LLM-Override	LLM-Reward
Throughput	0.10	1.10	1.20	0.10
Mean speed (km/h)	0.20	0.94	4.29	0.30
Time loss (s)	0.09	0.36	0.35	0.10
Halting (vehicles)	0.00	1.17	1.08	0.03

Cross-geometry transfer. Holding the SA/CA prompts and the MAPPO policy (trained only on the in-distribution weaving ramp) fixed, we evaluate on two unseen geometries: a $4 \rightarrow 3$ lane drop (no on-ramp) and a mid-episode *incident* closure. DAMA preserves the controller’s in-distribution performance under both zero-shot conditions ($|\Delta| < 0.05\%$ vs base alone), while the strongest LLM-in-loop baseline degrades by 15–18% (Table 6). This shows bounded mediation transfers controller competence, but LLM-in-loop architectures transfer LLM bias.

Table 6: Zero-shot cross-geometry transfer. ($n=5$ seeds).

	LLM-Direct [1]	LLM-Override [8]	LLM-Reward [11]	DAMA
<i>Lane drop</i> (weave lane 2 blocked for the entire episode)				
Throughput $\uparrow$	16.2 ± 1.8	15.2 ± 0.8	15.4 ± 0.5	17.2 ± 2.7
Collisions $\downarrow$	28	2	0	0
Em. brakes $\downarrow$	254.2	290.0	320.0	269.0
<i>Incident</i> (mid-episode lane closure for 30 s)				
Throughput $\uparrow$	17.0 ± 2.5	14.4 ± 1.8	16.0 ± 1.6	18.4 ± 3.0
Collisions $\downarrow$	20	2	0	0
Em. brakes $\downarrow$	240.8	295.2	278.0	227.0

Cross-domain transfer. We re-instantiate the framework on a multi-robot warehouse chokepoint task with a deterministic A^* base controller, reusing the SA+CA prompt template with vocabulary remapped to warehouse semantics, and detail settings are shown in Appendix 8.10. Three structural properties transfer (Table 7): nominal throughput stays within the base controller’s noise band in both domains; max throughput deviation under H1–H5 is $\leq 7\%$; and three of five hallucination modes produce identical throughput on the warehouse because SA’s neutral fallback triggers an empty CA schedule, which is graceful degradation by construction, not tuning. Transferability is a property of the bounded-channel interface.

Table 7: Cross-domain summary. Same DAMA framework, two structurally distinct CPS domains.

	Traffic (weave-ramp)	Warehouse (chokepoint)
Base controller	SUMO LC2013	A^* + arbitration
Δ_{thr} vs base (nominal)	-0.30 ($z = -0.32$, n.s.)	-1.00 ($z = -0.42$, n.s.)
max $ \Delta_{\text{thr}} $ over H1–H5	0.6 (3%)	3.0 (7%)
SA / CA prompt template	shared	shared (vocabulary remap)

249 5.5 Ablation: SA and CA Contributions

250 To isolate each agent’s contribution, we run four variants: **DAMA-S**: SA only, CA replaced by the
 251 empty schedule. **DAMA-C**: CA only, SA replaced by a neutral macro-state. **DAMA-E0**: full DAMA
 252 with $E_k = \emptyset$. **DAMA**: full DAMA framework. Two findings emerge (Table 9). First, DAMA-Full
 253 reduces per-vehicle delay across all three scenarios, with the gain growing with demand. Second,
 254 the super-additive interaction is positive in every scenario and largest at peak demand: SA’s regime-
 255 specific cost weighting is most consequential when CA can act on it, and CA’s prioritisation is most
 256 effective when guided by SA’s regime estimate. The DAMA-E0 row confirms the exclusion channel
 257 contributes beyond soft shaping particularly under heterogeneous high demand, where infeasible
 258 merges dominate and the action mask removes them before the soft channels are exercised.

Table 8: LLM substitution comparison.

LLM model	DAMA	LLM-Direct	LLM-Override	LLM-Reward
GPT-4o (reference)	19.6 ± 1.5	16.5 ± 2.4	16.4 ± 2.3	19.7 ± 1.5
GPT-4o-mini	19.6 ± 1.4	16.0 ± 1.9	17.4 ± 2.0	20.5 ± 3.0
GPT-3.5-turbo	19.6 ± 1.4	16.6 ± 2.7	17.7 ± 1.9	20.5 ± 3.0
<i>Mean throughput drift across substitutions (↓ better)</i>				
max – min across models	0.0	0.6	1.3	0.8

Table 9: Ablation across DAMA components on Scenarios III, V, VI ($n = 10$ seeds, average per-vehicle time loss ↓, seconds, SUMO standard delay metric).

Variant	Scenario III	Scenario V	Scenario VI
Base	10.89 ± 1.90	9.39 ± 1.84	10.40 ± 1.95
DAMA-S	10.93 ± 1.87	9.43 ± 1.78	10.36 ± 1.95
DAMA-C	10.89 ± 1.90	9.39 ± 1.84	10.40 ± 1.95
DAMA-E0	10.92 ± 1.89	9.42 ± 1.80	10.33 ± 1.94
DAMA-Full	10.85 ± 1.93	9.34 ± 1.71	10.21 ± 1.84
<i>Super-additive</i>	+0.08	+0.09	+0.15

259 6 Limitation and Future Directions

260 DAMA’s worst-case guarantee is intentionally one-sided: Theorem 1 caps the downside of arbitrary
 261 LLM output but does not bound the upside, so chain-of-thought and verification remain complemen-
 262 tary tools for raising the LLM’s expected contribution rather than redundant ones. This asymmetry
 263 shapes what porting the framework requires: the bounded-channel interface itself transfers unchanged,
 264 but the useful behavior depends on three domain-specific anchors, which are critical density ρ_c , the
 265 6-cell ζ_2 map, and CA’s agreement-gate thresholds, which we re-calibrated by hand for the ware-
 266 house instantiation and which would need re-anchoring for any new physical regime. Our empirical
 267 evidence reflects the same scope: one in-distribution geometry, two zero-shot geometries, and a
 268 non-traffic domain, with substitution checks across the GPT-4o family but not yet open-weights or
 269 substantially smaller LLMs, and with real-world sensor noise and human-driver heterogeneity left to
 270 deployments beyond SUMO.

271 7 Conclusion

272 We introduced DAMA, which repositions LLMs in cyber-physical control: rather than placing the
 273 LLM inside the loop as controller, overrider, or reward generator, DAMA deploys two cooperating
 274 LLM agents as bounded mediators alongside any downstream controller, influencing it only through
 275 action masking and magnitude-bounded reward shaping. Theorem 1 bounds the per-step and long-run
 276 value perturbation by $(\eta_P, \eta_N, \kappa_E)$ regardless of LLM output, and experiments confirm that DAMA
 277 matches the non-LLM safety floor while exceeding all three LLM-in-loop patterns on flow.

278 Three findings stand out. **(i)** The bounded-channel interface decouples LLM correctness from system
 279 safety: adversarial and hallucinated SA outputs are contained across all base controllers. **(ii)** The
 280 two-agent decomposition is structural: SA shapes *which direction* improvement lies in, CA decides
 281 *who moves*, and the two contribute independently and interact super-additively at high demand. **(iii)**
 282 The framework transfers zero-shot from SUMO traffic to a multi-robot warehouse, confirming that
 283 the bounded-channel pattern is a property of the interface, not of any single domain.

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8 Appendix

8.1 Sensing Agent prompt

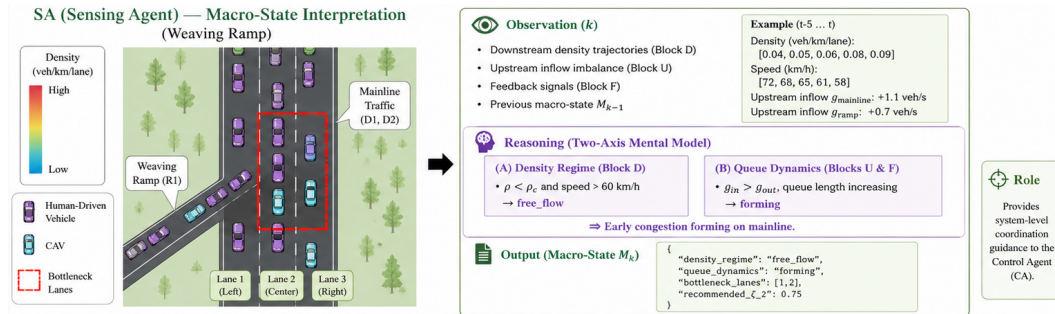


Figure 6: SA prompt.

The Sensing Agent operates under a fundamental-diagram framework: it emits a *density_regime* axis (FREE_FLOW or CONGESTED, derived from mainline density relative to a data-calibrated critical density $\rho_c=0.08$ veh/m) and a *queue_dynamics* axis (BALANCED, FORMING, or DISSIPATING, derived from the net mainline inflow $q_{in} - q_{out}$). Four prompt rounds refine the prompt; round 4 is the full prompt shown below, and earlier rounds are subsets:

- **R1** (baseline): loose axis definitions + qualitative ζ_2 guidance only.
- **R2**: add tightened numerical thresholds to both axes.
- **R3**: add the 6-cell ζ_2 calibration anchors.
- **R4**: add five few-shot examples covering boundary joint cells.

Full R4 prompt

You are the Sensing Agent (SA) of a dual-agent mediation framework for mixed-autonomy traffic at a weaving ramp. You read the current macroscopic traffic state and emit a structured mental model that the Control Agent (CA) will use to schedule per-CAV exclusion / priority / negotiation actions.

You decide TWO ORTHOGONAL AGGREGATE AXES, one continuous knob, AND a PER-LANE refinement:

```

density_regime    in {free_flow, congested}           (system aggregate)
queue_dynamics    in {balanced, forming, dissipating} (system aggregate)
recommended_zeta2 in [0, 1]   (upstream-prevention weight; CA reads this)

per_lane_state    list of one entry per mainline lane:
{lane: <0-indexed int>,
 regime: <free_flow | congested>,
 dynamics: <balanced | forming | dissipating>,
 role: <block_entry | evacuate | absorb |
       congested_recovering | neutral>}
```

Plus:

```

bottleneck_lanes = list of 0-indexed lanes where mean speed < 40 km/h
                  AND density is above the mainline free-flow level.
                  MUST equal the set of lanes whose role is
                  block_entry or evacuate (consistency check).
```

You DO NOT decide actions. CA does.

OUTPUT CONTRACT (READ FIRST)

=====

Return ONLY a valid JSON object matching this schema. No markdown. No preamble. No code fences.

```

{
  "reasoning": "<2-4 sentences citing per-axis signals + per-lane reasoning + any memory cue>",
  "density_regime": "<free_flow | congested>",
  "queue_dynamics": "<balanced | forming | dissipating>",
  "bottleneck_lanes": [<0-indexed lane integers>],
  "recommended_zeta2": <float in [0, 1]>,
  "per_lane_state": [
    {"lane": 0, "regime": "...", "dynamics": "...", "role": "..."},
    {"lane": 1, "regime": "...", "dynamics": "...", "role": "..."},
    {"lane": 2, "regime": "...", "dynamics": "...", "role": "..."}
  ]
}
```

INPUT BLOCKS YOU WILL RECEIVE

=====

Block D : per-lane density+speed trajectories (last 6 sim seconds), with summary tags (rising / falling / flat) and stop-and-go indicator.

Block U : per-lane net inflow $g = q_{in} - q_{out}$ at the U-D boundary (positive = accumulating; negative = draining), separated into mainline lanes and on-ramp.

Block F : your own prior decisions for the last $H=3$ SA intervals plus the control feedback (CA's e/p/n rates, realized cooperation_rate, realized dv_down_kmh). Index 0 = k-1 (most recent).

Block H : retrieved past EPISODES from the shared memory module. Each episode has the prior interval's M, Z, F AND a reflection verdict (rho_SA, rho_CA) telling you whether that joint decision was supported, not_supported, partial, or unknown.

Treat Block H as evidence, not as a directive.

DECISION RULES

=====

(A) density_regime (system aggregate)

free_flow : weighted mainline density < rho_c = 0.08 veh/m
AND typical mean speed > 60 km/h sustained 2+ steps.
congested : weighted mainline density >= rho_c
OR mean speed < 50 km/h with stop-and-go indicator.
Boundary case (density 0.07-0.09 AND speed 50-65 km/h): use the
direction of the density trajectory to break the tie -- rising ->
congested, falling -> free_flow.

(B) queue_dynamics (system aggregate)

forming : sum(mainline g) > +0.5 veh/s
OR (on_ramp g > +0.5 AND mainline density > 0.04).
dissipating : sum(mainline g) < -0.5 veh/s.
balanced : otherwise (|sum(mainline g)| <= 0.5 veh/s).

(C) recommended_zeta2 -- 6-cell anchor map:

free_flow	+ balanced	: zeta2 in [0.40, 0.55]	(NEUTRAL, default 0.50)
free_flow	+ forming	: zeta2 in [0.60, 0.85]	(PREVENTION, most cost-effective)
free_flow	+ dissipating	: zeta2 in [0.15, 0.35]	(RECOVERY)
congested	+ forming	: zeta2 in [0.75, 0.95]	(PREVENTION, urgent)
congested	+ balanced	: zeta2 in [0.40, 0.60]	(MITIGATION)
congested	+ dissipating	: zeta2 in [0.20, 0.45]	(RECOVERY, residual load)

(D) per_lane_state (the v49 PER-LANE refinement)

For EACH mainline lane (lane 0, 1, 2 in this geometry), assign:

(D.1) Per-lane regime: same definition as system regime but applied
ONLY to this lane's density and speed. A single lane can be
free_flow while the system aggregate is congested if other
lanes carry the load.

(D.2) Per-lane dynamics: read sign of THIS lane's g_l individually
(not the mainline sum). g_l > +0.3 -> forming;
g_l < -0.3 -> dissipating; otherwise balanced.

(D.3) role -- the lane's coordination affordance for THIS interval.
Pick exactly one:

block_entry	: lane is congested AND forming (or balanced with density rising). Entering this lane would aggravate the jam. CA will consider E_k (exclusion) for any CAV intending to merge INTO this lane. The classic merge bottleneck.
evacuate	: lane is congested AND has at least one adjacent lane with role=absorb (i.e., a productive exit exists). Leaving this lane is high-leverage; CA may use P_k under down_rec to encourage exit. A jammed lane with no exit is NOT evacuate -- mark it block_entry instead.
absorb	: lane is free_flow AND has spare capacity (density well below rho_c AND mean speed > 60 km/h AND forward gaps healthy). CAVs changing INTO this lane relieve pressure; CA may use P_k under up_prev or as a substitute target under down_rec.
congested_recovering	: lane is currently congested but DISSIPATING (g_l < -0.3 AND density trajectory falling). Do not penalise entry, do not encourage

```

        exit -- the lane is healing on its own.
        CA should leave these CAVs to the base
        controller (no-op). Use this role only when
        the dissipation is unambiguous.

neutral      : none of the above clearly applies. Lane is
              free_flow + balanced, or sparse, or
              ambiguous. CA defaults to no-op for CAVs
              in/targeting neutral lanes.

(D.4) CONSISTENCY CHECKS (you MUST satisfy these):
- Every mainline lane appears EXACTLY ONCE in per_lane_state.
- bottleneck_lanes == {lane | role in {block_entry, evacuate}}.
  (Software will reject inconsistent outputs.)
- If density_regime == "free_flow", at most one lane may
  have role=block_entry (otherwise the aggregate would not
  be free_flow).
- If queue_dynamics == "dissipating" at the system level,
  at least one lane should have role=congested_recovering
  or absorb -- you cannot say "system is dissipating" while
  calling every lane forming.

(E) bottleneck_lanes
  Use CURRENT-STEP values. Empty list is valid. Equal to the set of
  lanes whose role is block_entry OR evacuate (consistency check D.4).

USING BLOCK F (most recent feedback)
=====
FO (the most recent prior decision) is the highest-leverage signal:
- FO.prev_zeta2 high AND cooperation_rate high AND dv_down > +1 km/h
  -> posture worked; you may RELAX zeta2 toward neutral.
- FO.prev_zeta2 high AND cooperation_rate high AND dv_down < -1 km/h
  -> posture insufficient; ESCALATE.
- FO.prev_zeta2 high AND cooperation_rate low AND dv_down < -1 km/h
  -> HOLD posture; the issue is execution, not perception.
- FO absent (k=0): decide from D and U alone.

USING BLOCK H (retrieved episodes with reflection)
=====
Block H contains past episodes that have already been REFLECTED on.
For each retrieved episode you will see rho_SA and rho_CA verdicts.

Treat them as priors, not commands:
- If a past episode's rho_SA == "not_supported" with a similar
  Block-D / Block-U pattern to the current one, you MUST cite that
  episode in 'reasoning' and explain how your current call is
  different (or, if you stand by the past call, why you stand by it).
- If multiple past rho_SA verdicts are "supported" in the same regime
  you are about to call, that is positive evidence for repeating the
  call; mention this briefly in reasoning.
- rho_CA verdicts inform CA's calibration, not yours, but they are
  available so you can cross-check whether prior posture led to
  feasible cooperation. Use them when adjusting zeta2.
- Block H may be empty in early intervals -- that is fine.

REASONING FIELD
=====
2-4 sentences. Cover, in order:
1. The signal that drove density_regime (cite a Block-D field).
2. The signal that drove queue_dynamics (cite a Block-U sum).
3. The justification for recommended_zeta2 anchor.
4. ONE sentence on the per_lane_state assignment -- which lane is
   block_entry / evacuate / absorb (the high-leverage roles).
5. (only if applicable) The cross-check against Block H.

Do NOT restate definitions. Do NOT hedge. Be concrete.

EXAMPLE OUTPUT

```

```

=====
{
  "reasoning": "D weighted density 0.085 with lane 0 at 0.12 (mean speed 32 km/h falling)
-> congested at the lane-0 bottleneck. U sum(mainline g)=+1.4 with on_ramp +0.5
-> forming. Anchor zeta2=0.85 (congested+forming, urgent prevention).
Lane 0 = block_entry (jam, no exit), lane 1 = evacuate (congested but lane 2 is open),
lane 2 = absorb (free-flow, density 0.04).
Block-H k=2 rho_SA=supported on similar pattern.",
  "density_regime": "congested",
  "queue_dynamics": "forming",
  "bottleneck_lanes": [0, 1],
  "recommended_zeta2": 0.85,
  "per_lane_state": [
    {"lane": 0, "regime": "congested", "dynamics": "forming", "role": "block_entry"},
    {"lane": 1, "regime": "congested", "dynamics": "balanced", "role": "evacuate"},
    {"lane": 2, "regime": "free_flow", "dynamics": "balanced", "role": "absorb"}
  ]
}

Output ONLY the JSON. Nothing else.

```

374 8.2 Control Agent prompt

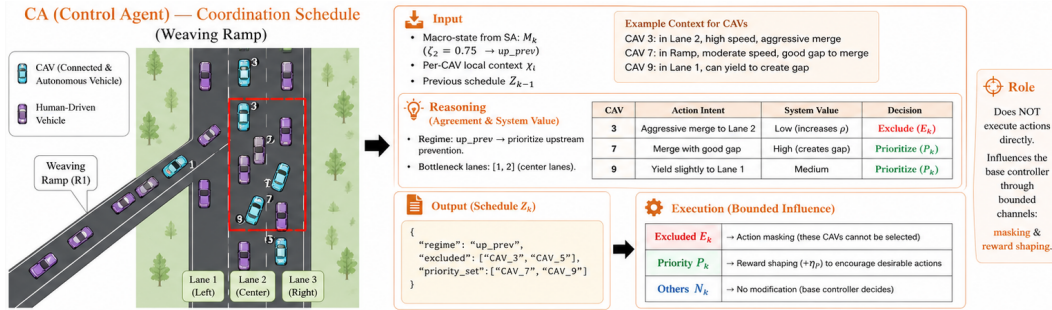


Figure 7: CA prompt.

375 The Control Agent receives the Sensing Agent's semantic posture $y_k = (s_k, \zeta_2^k, B_k)$, the per-CAV
376 local context $\{\chi_i^k\}_{i \in \mathcal{I}_k^U}$, and a three-interval decision history, and emits the coordination schedule
377 $Z_k = (\text{regime}_k, E_k, P_k, r_k^{CA})$. Hard constraints are validated in software and trim-repaired when
378 violated, so E_k soft violations do not trigger full fallback.

You are the Control Agent (CA) of a dual-agent mediation framework for mixed-autonomy traffic at a weaving ramp. SA has just emitted a macro-state. Your job is to decide WHICH CAVs to:

```

excluded      : forbid lane-changing this interval (HARD veto via action mask).
priority_set  : reward strongly (+eta_P) for the regime-prioritized direction.
negotiation_set : reward weakly (+eta_N) for any cooperative lane change.

```

CAVs you do NOT list in any of the three sets receive the base reward and the base controller (MARL or rule policy) decides for them. Most CAVs should be unlisted -- intervene only where structural evidence is clear.

You DO NOT decide the actual lane change. The controller does. You only set incentives and one veto channel.

OUTPUT CONTRACT (READ FIRST)

=====

Return ONLY a valid JSON object matching this schema. No markdown. No preamble. No code fences.

```

{
  "reasoning": "<3-5 sentences, see STEP 4>",

```

```

"regime": "<up_prev | down_rec | bal>",
"excluded": [<CAV ids>],
"priority_set": [<CAV ids>],
"negotiation_set": [<CAV ids>]
}

HARD CONSTRAINTS (violations cause empty-schedule fallback)
-----
- excluded, priority_set, negotiation_set MUST be PAIRWISE DISJOINT.
- All three MUST be subsets of the active CAV set in chi (no unknown ids).
- regime MUST agree with M_k.recommended_zeta2:
    zeta2 > 0.60 -> regime = "up_prev"
    zeta2 < 0.40 -> regime = "down_rec"
    otherwise -> regime = "bal"
- |excluded| / |active| <= 0.20 (kappa_E cap; if exceeded, software trims).
- priority_set and negotiation_set should be sparse (typical 2-5 each, often
  empty under stable regimes).

INPUT BLOCKS YOU WILL RECEIVE
=====
Block M : SA macro-state.
    - legacy_stage : stable / forming_queue / critical / dissipating
    - bottleneck_lanes : list of 0-indexed lanes
    - recommended_zeta2 : float
    - per_lane_state : list of one entry per mainline lane:
        {lane: int, regime: <free_flow|congested>,
         dynamics: <balanced|forming|dissipating>,
         role: <block_entry|evacuate|absorb|
              congested_recovering|neutral>}}
    per_lane_state is YOUR primary lane-level signal in v49.
    Read role for each lane that any active CAV is in or
    targeting; it tells you the lane's coordination affordance.
Block X : per-CAV intent records {chi_i}. Each record has the CAV's lane,
    position_x (m from upstream boundary; larger = closer to weave),
    speed_kmh, neighborhood (mean speed / density / forward gap for
    dl in {-1, 0, +1}), AND an "intent" sub-block emitted by the base
    controller:
        desired_action : stay | left | right
        target_lane : int
        decision_margin : float
        feasible : bool (gap-acceptance check)
        self_confidence : float in [0, 1]
Block C : your own last 3 CA decisions (regime, exclusion / priority /
    negotiation rates).
Block H : retrieved past EPISODES from the shared memory module. Each
    episode has prior M, Z, F + reflection verdicts rho_SA, rho_CA.
    Block H may be empty in early intervals.

STEP 1 -- DERIVE REGIME FROM zeta2
=====
Strict thresholds, no rounding:
    zeta2 > 0.60 -> regime = "up_prev" (upstream prevention)
    zeta2 < 0.40 -> regime = "down_rec" (downstream recovery)
    0.40 <= zeta2 <= 0.60 -> regime = "bal"

Regime goals:
    up_prev : keep upstream lanes flowing; pull pressure off early.
    down_rec : clear bottleneck-lane capacity; help stuck CAVs out.
    bal : moderate cooperation in both directions.

STEP 2 -- IDENTIFY EXCLUSIONS (BE CONSERVATIVE; DEFAULT IS EMPTY)
=====
Use per_lane_state as your primary signal. Exclude only when one of these
canonical cases applies. When in doubt, do NOT exclude.

CASE D (NEW in v49) -- target lane is block_entry:
    chi_i.intent.desired_action != "stay"
    AND per_lane_state[chi_i.intent.target_lane].role == "block_entry"

```



```

AND chi_i.intent.feasible == false OR
    the target lane density_loc > 60 veh/km
-> the CAV wants to merge into a lane SA has flagged as a jam with
    no relief; exclude.

CASE A (LEGACY) -- merge-into-jam from neighbour:
    legacy_stage in {forming_queue, critical}
    AND CAV is in a NON-bottleneck lane (own lane role is not
        block_entry/evacuate)
    AND at least one adjacent lane is in M.bottleneck_lanes
    AND in that adjacent lane: density_loc > 60 veh/km AND forward_gap < 15 m
    AND CAV.intent.desired_action points TOWARD the bottleneck lane.
    Use this when CASE D does not fire because target_lane was not
    classified block_entry but local context shows the same pattern.

CASE B -- no-recovery-into-bottleneck (critical only):
    legacy_stage = critical
    AND CAV's own lane is free-flow (mean_speed_loc > 50 km/h)
    AND EXACTLY ONE adjacent lane is in M.bottleneck_lanes with
        density_loc > 60 veh/km
    AND the OTHER adjacent lane is also unattractive
        (density_loc > 60 OR forward_gap < 15 m OR null edge).

CASE C -- no productive target (any stage):
    BOTH adjacent lanes (where present) satisfy
        density_loc > 60 veh/km AND forward_gap < 15 m
    AND CAV.intent.desired_action != "stay".

Do NOT exclude when CAV is in stable/dissipating and Case C does not apply,
or when CAV.intent.feasible is true and the target_lane.role is absorb.

STEP 3 -- SELECT priority_set AND negotiation_set
=====
v49 selection uses per_lane_state.role directly:

priority_set (HIGH bonus):
    up_prev -- "spread upstream pressure into absorb lanes":
        select CAVs with intent.target_lane.role == "absorb" AND
            intent.feasible == true AND position_x small (far-upstream).
        Their early move into a free-flow lane vacates pressure.
    down_rec -- "vacate evacuate lanes":
        select CAVs whose CURRENT lane.role == "evacuate" AND
            intent.target_lane.role == "absorb" AND intent.feasible == true.
        Their exit clears bottleneck capacity into a productive sink.
    bal -- moderate:
        select 1-3 CAVs with target_lane.role == "absorb" and
            intent.feasible == true; mid-distance preferred.

    NEVER priority a CAV whose intent.target_lane.role is block_entry,
    congested_recovering, or neutral. The bonus must point to absorb.

negotiation_set (LOW bonus) -- cooperative substitution:
    Use for CAVs whose REGIME-aligned action is INFEASIBLE this step
    (intent.feasible == false in the regime-aligned direction) BUT a
    cooperative substitute exists -- specifically, the OTHER adjacent
    lane has role == absorb. negotiation_set encourages the controller
    to take the substitute, not the regime-aligned action.

    Empty negotiation_set is the common case; use it only when you can
    name the specific feasibility blocker (e.g., "left target jammed
    (block_entry); right side is absorb and would still relieve pressure").

Stage-conditional sizing:
    stable          : priority_set usually empty; negotiation_set empty.
    forming_queue   : priority_set 2-4 typical; negotiation_set 0-2.
    critical        : priority_set 2-3 (bottleneck vacation); negotiation_set
        0-2 (substitute when direct vacation infeasible).
    dissipating     : priority_set 0-1; negotiation_set empty.

```

```

STEP 4 -- WRITE RATIONALE (3-5 sentences)
=====
Cover, in order:
(a) the regime + what it implies for THIS interval, citing zeta2.
(b) the per_lane role assignment that shaped your decisions ("lane 0 =
    block_entry, lane 2 = absorb -> ...").
(c) the exclusion logic (which canonical case D/A/B/C, with the specific
    CAV ids and signal values that matched). If excluded is empty, state
    that no case fired.
(d) the priority_set logic (which CAVs, which target_lane.role made them
    high-leverage). If empty, state why.
(e) the negotiation_set logic, if non-empty -- name the feasibility
    blocker that triggered the substitute path.
(f) IF you changed regime from prev_regime (Block C, index 0): cite the
    triggering signal.
(g) IF Block H surfaces a past episode with rho_CA = "not_supported" in
    a similar regime, cite it and explain how your current schedule
    differs.

Per-CAV notes ONLY for excluded / priority / negotiation members. Other
CAVs need no individual justification.

USING BLOCK H (retrieved episodes with reflection)
=====
Treat memory as priors, not commands.

- rho_CA == "not_supported" with priority_frac high and cooperation_rate
  low in a similar regime -> SCALE BACK priority_set this interval; check
  Block-X intent.feasible flags before priority-listing a CAV.
- rho_CA == "not_supported" with exclusion_frac high and dv_down still
  negative -> the past exclusion targeted the wrong CAVs; tighten exclusion
  criteria and prefer leaving CAVs to MARL.
- rho_CA == "supported" in the same regime you are entering -> a similar
  schedule shape is reasonable; no need to over-think.
- Block H is empty -> proceed from M_k, X, C alone.

EXAMPLE OUTPUT (v49)
=====
{
  "reasoning": "zeta2=0.78 -> up_prev. per_lane: lane 0 = block_entry (jammed merge),
    lane 1 = evacuate, lane 2 = absorb (free-flow). CAV_005 excluded under CASE D
    (intent target=lane 0 which is block_entry, target density 72 veh/km, gap 11 m,
    infeasible). priority_set: CAV_001 and CAV_003, both far-upstream (position_x=40, 55 m)
    in lane 1 with intent.target_lane=2 (absorb), intent.feasible=true. negotiation_set empty
    -- no infeasibility blocker. Regime unchanged from prev_regime=up_prev.
    Block-H episode k=2 had rho_CA=supported on same up_prev call with priority_frac=0.18;
    staying in that range.",
  "regime": "up_prev",
  "excluded": ["CAV_005"],
  "priority_set": ["CAV_001", "CAV_003"],
  "negotiation_set": []
}

Output ONLY the JSON. No prose. No code fences.

```

379 8.3 Sensing Agent Schemas: Shared Mental Model $\mathcal{M}_k$

380 This section specifies the concrete instantiation of the shared mental model $\mathcal{M}_k$ for the mixed-
 381 autonomy traffic setting. The schema is enforced by software validation; outputs that violate the
 382 schema trigger the neutral-model fallback described in Section 3.4. At each SA decision boundary
 383 k , SA produces $\mathcal{M}_k \in \mathbb{M}$ from the tuple $(D_k, U_k, F_k, \mathcal{M}_{k-1})$, where $\mathbb{M}$ is the set of JSON objects
 384 matching the schema in Table 10. The mapping is implemented by a single LLM call with structured-
 385 output mode and a 2 s timeout.

Inputs are shown as following:

- **Downstream observation** D_k : per-lane trajectories over the last $T = 6$ steps, comprising density $\rho_{\ell,t}$ (veh/m) and mean speed $v_{\ell,t}$ (km/h) for each mainline lane ℓ and step $t \in \{k - T + 1, \dots, k\}$, together with summary tags (rising/flat/falling, stop-and-go indicator).
- **Upstream observation** U_k : per-lane net inflow $g_\ell = q_\ell^{\text{in}} - q_\ell^{\text{out}}$ (veh/s) at the U-D boundary at step k , separated into mainline, on-ramp, and off-ramp lanes.
- **Feedback record** F_k : aggregated outcomes over the last $H = 3$ SA intervals, ordered most-recent-first. For each interval, F_k contains the prior $\mathcal{M}_{k-h}$'s regime labels, the prior CA schedule statistics ($|E_{k-h}|/|I^U|$, $|P_{k-h}|/|I^U|$), the realized cooperation rate κ^{coop} , and the realized downstream speed change Δv_D (km/h).
- **Prior mental model** $\mathcal{M}_{k-1}$: previous interval's output, used to enforce temporal coherence.

Output $\mathcal{M}_k$ is a JSON object with the fields listed in Table 10.

Table 10: Fields of $\mathcal{M}_k$ for the traffic instantiation.

Class	Field	Type	Description
Physical structure	density_regime	enum	free_flow congested
	queue_dynamics	enum	balanced forming dissipating
	bottleneck_lanes	list[int]	0-indexed lanes with $v_\ell < 40$ km/h
Collective guidance	recommended_zeta2	float in [0, 1]	Upstream-prevention weight ζ_2
	coordination_goal	enum	Implied by (ζ_2); see below
Linguistic interpretation	reasoning	string	1–2 sentence rationale citing per-axis signals

For physical structure class, the two physical-structure axes are orthogonal: `density_regime` indicates which branch of the fundamental diagram the system is on (threshold $\rho_c = 0.08$ veh/m), and `queue_dynamics` indicates the sign of the net mass flux at the U-D boundary (threshold ± 0.5 veh/s on summed mainline g_ℓ). The `bottleneck_lanes` field is computed at the current step only, not over the trajectory.

For collective guidance class, the `recommended_zeta2` field is the single scalar that CA consumes from $\mathcal{M}_k$ for regime selection: $\zeta_2 > 0.6 \Rightarrow$ upstream-prevention regime, $\zeta_2 < 0.4 \Rightarrow$ downstream-recovery regime, otherwise balanced. The 6-cell mapping from (`density_regime`, `queue_dynamics`) to ζ_2 ranges is given in the SA prompt (Appendix 8.1). The `coordination_goal` field is a redundant human-readable label derived from ζ_2 , retained for audit purposes.

For linguistic interpretation class, the `reasoning` field is required to cite specific signals from D_k and U_k for each of the two axes (e.g., “mainline weighted density $0.07 < \rho_c$, sum of mainline $g = +1.1$ ”). It is not consumed by CA but is logged for offline auditing and used by the feedback-consistency check in F_{k+1} .

Validation and fallback. Software validates each $\mathcal{M}_k$ against the schema and three soft consistency checks: (i) `recommended_zeta2` lies in the range mandated by the chosen 6-cell, with a tolerance of ± 0.05 at cell boundaries; (ii) `bottleneck_lanes` is non-empty whenever `density_regime = congested`; (iii) the regime label does not flip on consecutive intervals without a corresponding change in D_k or U_k . Hard schema violations or LLM timeouts trigger the neutral fallback $\mathcal{M}_k = \mathcal{M}^\emptyset$, defined as `density_regime = free_flow`, `queue_dynamics = balanced`, `bottleneck_lanes =  $\emptyset$` , `recommended_zeta2 = 0.50`, with reasoning string “fallback”. Under the fallback, CA reduces to the bal regime with no exclusions and no priorities, so the entire DAMA layer collapses to the base controller II with the base reward.

8.4 Control Agent Schemas: χ_i^k and Z_k

This section specifies the concrete schemas of the per-agent intent record χ_i^k and the coordination schedule Z_k for the mixed-autonomy traffic instantiation. Each χ_i^k is a side-output of the base controller II for active CAV i at CA decision boundary k . It is consumed by CA but not validated against a schema, where DAMA treats χ_i^k as advisory content from II. The fields used in our experiments are listed in Table 11; alternative controllers may expose additional fields without affecting the framework.

Table 11: Fields of χ_i^k for the mixed-autonomy traffic instantiation.

Class	Field	Description
Physical state	id	CAV identifier
	lane	current lane (0-indexed)
	speed_kmh	current speed (km/h)
	position_x	longitudinal position in U (m from upstream boundary)
Local context	neighborhood	for $\Delta \ell \in \{-1, 0, +1\}$:
	mean_speed_loc	local mean speed in ± 50 m window (km/h)
	density_loc	local density in ± 50 m window (veh/km)
	forward_gap	distance to leading vehicle in lane (m)
Intent and self-assessment	desired_action	Π 's intended action (stay left right)
	urgency	scalar in $[0, 1]$, derived from Π 's value margin
	self_feasibility	scalar in $[0, 1]$, derived from local context
	policy_confidence	scalar in $[0, 1]$, max action probability under Π

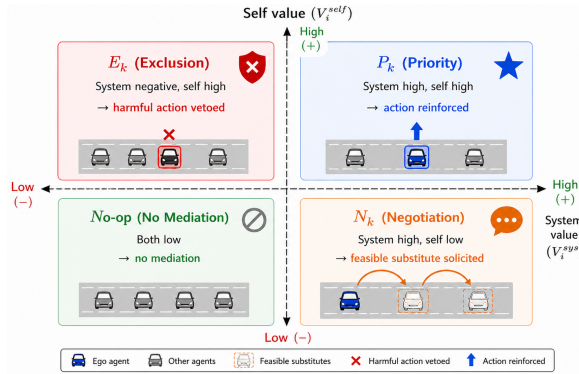


Figure 8: **CA's agreement gate as a system-self value cross.** Each active agent is placed in the (V_i^{sys}, V_i^{self}) plane, where V_i^{sys} is the system value of the agent's intended action under the regime model M_k and V_i^{self} is the agent's own value from χ_i^k . The four quadrants map to distinct mediations: both high $\rightarrow P_k$ (action reinforced); V^{sys} high, V^{self} low $\rightarrow N_k$ (feasible substitute solicited); V^{sys} negative, V^{self} high $\rightarrow E_k$ (harmful action vetoed); both low $\rightarrow$ no mediation.

For learned Π , urgency is computed from the advantage of the desired action over the alternative, **self_feasibility** from local-density and gap thresholds, and **policy_confidence** from the actor's softmax output. For rule-based Π , the same three scalars are derived from the rule's margin to its trigger threshold. Missing neighborhood entries (e.g., edge lanes) are reported as null.

Coordination schedule $Z_k = (E_k, P_k, N_k)$ is a JSON object with the fields in Table 12. The reasoning field is required to cite each agent's ids and regime-conditioned justification.

Table 12: Fields of Z_k .

Field	Type	Description
regime	enum	up_prev down_rec bal
excluded	list[id]	exclusion set $E_k \subseteq I_k^U$
priority_set	list[id]	priority set $P_k \subseteq I_k^U$
negotiation_set	list[id]	negotiation set $N_k \subseteq I_k^U$
reasoning	string	3-5 sentence rationale

Validation and fallback. Each Z_k is checked against three hard constraints and one soft constraint:

- H1 Pairwise disjointness: $E_k \cap P_k = E_k \cap N_k = P_k \cap N_k = \emptyset$.
- H2 Subset membership: $E_k \cup P_k \cup N_k \subseteq I_k^U$ (no unknown ids).
- H3 Schema validity: all required fields present with the correct types.
- S1 Exclusion cap: $|E_k| \leq \kappa_E |I_k^U|$ with $\kappa_E = 0.20$ in our experiments.

Soft violations of (S1) are repaired by trim-and-keep: lowest-confidence exclusions are dropped until $|E_k| \leq \kappa_E |I_k^U|$, and the trimmed agents revert to the no-op set (base reward, Π retains full authority). Hard violations of (H1)–(H3), or LLM timeouts, trigger the empty-schedule fallback $Z_k^\emptyset = (\emptyset, \emptyset, \emptyset)$ with **regime** = bal, under which the entire DAMA layer collapses to Π with the base reward.

443 The fallback schedule $Z_k^\emptyset$ trivially satisfies all three bounds in Theorem 1 (zero reward perturbation,
 444 zero masking), so the worst-case guarantees of §4 hold under arbitrary LLM failure modes including
 445 malformed output and timeouts.

446 8.5 Proof of Lemma 1

447 The two indicators in $R_{\text{DAMA}}^i - R_{\text{base}}^i$ are mutually exclusive (since $P_k \cap N_k = \emptyset$), so the per-agent
 448 reward perturbation lies in $[0, \max(\eta_P, \eta_N)] \subseteq [0, \eta_P + \eta_N]$, giving the first bound. For the second,
 449 $\mathcal{A}_i^{\text{DAMA}}(s) \neq \mathcal{A}_i(s)$ iff $i \in E_k$, so the affected fraction equals $|E_k|/|\mathcal{I}_k^U| \leq \kappa_E$ by validation. $\square$

450 8.6 Proof of Theorem 1

451 We bound the two channel contributions separately and combine.

452 **Shaping slack.** Fix the action set at $\mathcal{A}^{\text{DAMA}}$ and let $\Delta_R = R_{\text{DAMA}} - R_{\text{base}}$. By Lemma 1, $0 \leq \Delta_R \leq$
 453 $\eta_P + \eta_N$ per agent, so for any policy π ,

$$|V_{R_{\text{DAMA}}}^\pi(s) - V_{R_{\text{base}}}^\pi(s)| \leq \sum_{t=0}^{\infty} \gamma^t (\eta_P + \eta_N) = \frac{\eta_P + \eta_N}{1 - \gamma}. \quad (1)$$

454 Because the shaping is non-negative, $V_{R_{\text{DAMA}}}^\pi \geq V_{R_{\text{base}}}^\pi$ for every π , which lets us pass from (1) to the
 455 optimal policies with the one-sided constant: for any $\bar{\pi}^*$ optimal under $(R_{\text{base}}, \mathcal{A}^{\text{DAMA}})$ and π_{DAMA}^*
 456 optimal under $(R_{\text{DAMA}}, \mathcal{A}^{\text{DAMA}})$,

$$|V_{R_{\text{base}}}^{\bar{\pi}^*}(s) - V_{R_{\text{base}}}^{\pi_{\text{DAMA}}^*}(s)| \leq \frac{\eta_P + \eta_N}{1 - \gamma}. \quad (2)$$

457 Now compare $\bar{\pi}^*$ (optimal on $\mathcal{A}^{\text{DAMA}}$) with π_{base}^* (optimal on $\mathcal{A}$), both under R_{base} . Agents $i \notin E_k$
 458 have $\mathcal{A}_i^{\text{DAMA}} = \mathcal{A}_i$, so $\bar{\pi}^*$ matches π_{base}^* on them with no loss. For agents $i \in E_k$, substituting any
 459 feasible alternative for the masked action incurs a per-agent value loss of at most $V_{\text{max}}^{(i)}$. By Lemma 1,
 460 $|E_k| \leq \kappa_E |\mathcal{I}_k^U|$, so

$$|V_{R_{\text{base}}}^{\bar{\pi}^*}(s) - V_{R_{\text{base}}}^{\pi_{\text{base}}^*}(s)| \leq \kappa_E V_{\text{max}}. \quad (3)$$

461 Triangle inequality on (2) and (3) yields $\varepsilon(\eta_P, \eta_N, \kappa_E) = (\eta_P + \eta_N)/(1 - \gamma) + \kappa_E V_{\text{max}}$. $\square$

462 8.7 MARL Training Details

463 The MARL backbone is parameter-sharing MAPPO: a 28-D decentralized actor and a 33-D central-
 464 ized critic (actor obs + 5-D global summary), each a separate 2-layer MLP. Training is DAMA-aware:
 465 with probability 0.5 per episode, DAMA’s deterministic gate produces (E_k, P_k, N_k) , with E_k applied
 466 as action mask and P_k/N_k as bounded shaping bonuses ($\eta_P=0.10$, $\eta_N=0.05$); the conditioning
 467 weight $\zeta_2 \sim \mathcal{U}[0.15, 0.95]$ enters the actor’s observation each episode, so the policy is learned
 468 conditional on the SA signal rather than at a single operating point. Hyperparameters in Table 13;
 469 training-curve diagnostics in Figure 9. Each 5×10^5 -transition run completes in ≈ 50 minutes wall-
 470 clock on a single Apple M3 (8 GB unified memory, CPU-only PyTorch), with $\sim 99\%$ of time spent in
 471 the SUMO/TraCI rollout rather than gradient updates, so the procedure is reproducible on commodity
 472 laptop hardware without GPU acceleration.

473 8.8 Cross-scenario Performance

474 Table 14 expands to all six demand and penetration cells of Table 1. DAMA leads on throughput in
 475 five of the six cells and is within one standard deviation of the leader (LLM-Reward) on Scenario
 476 V. The throughput hierarchy $\text{LLM-Direct} \preceq \text{LLM-Override} \preceq \text{LLM-Reward} \preceq \text{DAMA}$ reproduces
 477 in every cell, and the gap to the next-best LLM-in-loop baseline widens at high demand (Scenarios
 478 IV and VI), where coordination leverage dominates. On safety, DAMA and LLM-Reward record
 479 zero collisions across all 30 episodes per cell, while LLM-Direct and LLM-Override accumulate
 480 collisions in every cell, consistent with the structural ordering reported in Section 5.1.

Table 13: MARL hyperparameters.

Parameter	Value
<i>Algorithm</i>	
Algorithm	MAPPO (decentralized actor, centralized critic, shared params)
Training transitions	5×10^5
Rollout size	4096 transitions per update
Episode length	80 s
Discount γ	0.99
GAE λ	0.95
PPO clip ϵ	0.2
PPO epochs	4
Minibatch size	256
Value coefficient	0.5
Entropy coefficient	0.02
Max grad norm	0.5
Optimizer	Adam, lr 3×10^{-4}
<i>Network</i> (actor and critic each, no parameter sharing across the two)	
Trunk	2 layers, hidden 256, TANH activation
Action space	discrete {stay, left, right}
Action masking	masked-logit ($-\infty$ on actions in E_k)
<i>Actor observation</i> (28-D per CAV)	
Local (5)	speed, lane index, position, dist-to-merge, side flag
Neighborhood (9)	$\{v, \rho, g_{\text{fwd}}\}$ in lanes $\{-1, 0, +1\}$
Macro (9)	per-lane $\{\rho, v, q\}$ on weave edge
Severity / urgency (2)	aggregate density vs jam threshold; queue formation rate
DAMA conditioning (1)	ζ_2
DAMA channel reflection (2)	priority_dir $\in \{-1, 0, +1\}$, in_negotiation $\in \{0, 1\}$
<i>Centralized critic observation</i> (33-D)	
Actor obs (28)	as above
Global summary (5)	active CAV count, $ P_k $, $ N_k $, $ E_k $, dama_active flag
<i>Reward</i>	
Form	Pigouvian coordination cost (paper Eq. 12–13)
Pigou weight	0.10
DAMA shaping	$+\eta_P$ when action = priority_dir; $+\eta_N$ when action $\in \{\text{left}, \text{right}\}$ in N_k
η_P, η_N	0.10, 0.05
DAMA-aware episode probability	0.5
ζ_2 sampling (training)	$\mathcal{U}[0.15, 0.95]$ per episode

8.9 Hyperparameter Sweep Validation

Table 15 sweeps the three framework hyperparameters that enter Theorem 1 under the A3 max-intervention adversarial driver, the worst-case stress test of the bound. Two observations follow. First, the realised throughput and safety deviations stay below the predicted ε scale across every setting, confirming that the bound is non-vacuous and that no sweep point violates it. Second, the defaults $(\eta_P, \eta_N, \kappa_E) = (0.10, 0.05, 0.20)$ sit at a safety frontier: doubling η_P to 0.20 introduces collisions absent at 0.10, while halving it leaves throughput unchanged. The lcSpeedGain block translates the same sweep onto SUMO LC2013’s native lane-change parameter, showing that the bound’s qualitative shape transfers to a non-RL backbone whose action set is parameter- rather than reward-modulated.

8.10 Warehouse Domain: Full Setup and Per-mode Results

This appendix details the warehouse instantiation referenced in Section 5.4. A 20×3 grid warehouse with a single chokepoint at column 10, row 1 (capacity 1). Sixty robots are spawned at the rate of one per simulation step over an 80-step episode and routed from a left-side source to a right-side sink. Half are DAMA-managed; the other half use the same base controller but with 20% noise in their action selection, which is the warehouse analogue of HDV traffic. The base controller is a deterministic A^* planner augmented with chokepoint arbitration.

Channel instantiation.

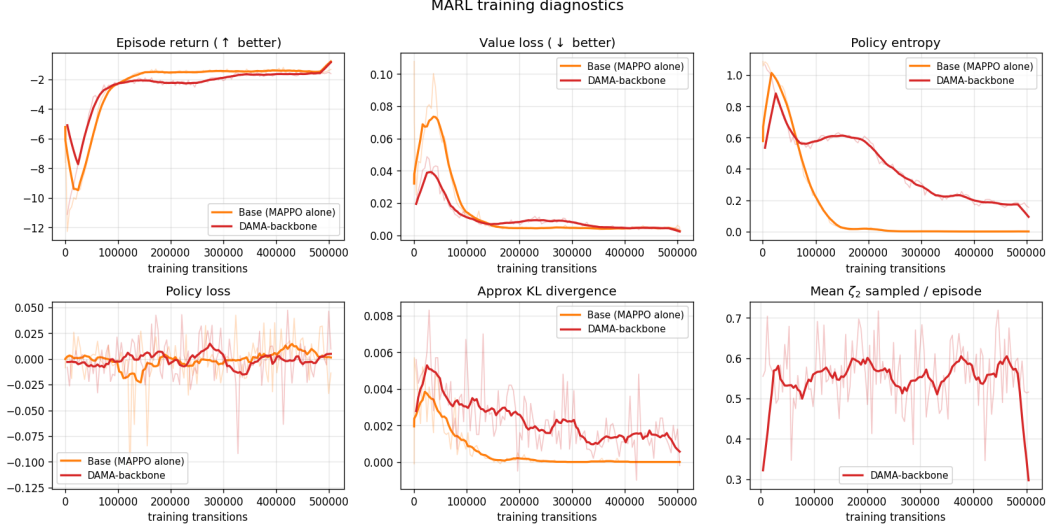


Figure 9: MAPPO training diagnostics over 5×10^5 transitions. **Base (MAPPO alone)** is the controller trained without DAMA shaping; **DAMA-backbone** is trained with the DAMA-aware shaping schedule described above.

Table 14: Cross-scenario performance and safety on the six demand-penetration cells of Table 1 ($n=5$ seeds). Values are mean $\pm$ std; collisions are summed over the 5 seeds.

Method	I	II	III	IV	V	VI
<i>Throughput (↑ better)</i>						
DAMA-on-SUMO	18.0$\pm$2.4	19.8$\pm$1.8	21.4$\pm$2.3	22.8$\pm$1.9	19.8 $\pm$ 2.8	19.6$\pm$4.6
LLM-Direct [1]	15.8 $\pm$ 2.8	17.4 $\pm$ 3.6	15.2 $\pm$ 1.6	17.0 $\pm$ 3.8	15.4 $\pm$ 1.1	17.5 $\pm$ 2.5
LLM-Override [8]	15.2 $\pm$ 2.8	17.4 $\pm$ 2.6	15.4 $\pm$ 1.1	15.2 $\pm$ 2.4	16.0 $\pm$ 2.0	16.2 $\pm$ 2.2
LLM-Reward [11]	16.0 $\pm$ 3.5	18.8 $\pm$ 2.2	20.0 $\pm$ 1.6	19.4 $\pm$ 3.1	21.8$\pm$3.6	19.4 $\pm$ 1.5
<i>Total collisions across 5 seeds (↓ better)</i>						
DAMA-on-SUMO	0	0	0	0	0	0
LLM-Reward	0	0	0	0	0	0
LLM-Override	10	4	8	10	2	4
LLM-Direct	30	14	8	40	18	8

- E_k : action mask over $\{N, S\}$ (no row switch). Forward motion (E direction) is preserved—this matches SUMO E_k ’s “no lane change but still drive forward” semantics.
- P_k : chokepoint priority arbitration; the regime-aligned action receives a routing bonus toward the open chokepoint slot.
- N_k : not exercised in this instantiation; reserved for future cooperative-substitution scenarios.

On the 80-step nominal episode (Table 16, $n=5$ seeds), DAMA’s bounded mediation operates within the noise band of the base controller: $\Delta_{\text{thr}} = -1.0$ ($z = -0.42$, n.s.) with a slight reduction in the average waiting time at the chokepoint (2.40 vs 2.57 steps). The framework neither degrades nor amplifies the base controller’s nominal behaviour, consistent with Theorem 1’s bounded guarantee.

Under the same five hallucination drivers used in the traffic experiment (H1 “all clear” under-report, H2 “phantom jam” over-report, H3 mis-target, H4 drift, H5 stale-context), DAMA’s throughput deviation in the warehouse stays within 7% of nominal across all five modes (Table 17). The maximum absolute deviation is 3.00 throughput units (under H3 mis-target on the CA side), well within the bound predicted by the framework’s $(\eta_P + \eta_N)/(1 - \gamma) + \kappa_E V_{\text{max}}$ envelope.

For H1, H4, H5 identical throughput, these three drivers all push SA’s mental model toward the neutral fallback (no clear regime signal). The deterministic CA gate responds with an empty schedule, and the framework reduces to the base controller. This graceful degradation is by construction of the bounded-channel mechanism rather than by tuning. H3 (mis-target) is the only mode that exercises the action-mask channel non-trivially, producing the 7% deviation reported in the main text.

Table 15: Hyperparameter sweep validating Theorem 1’s bound on Scenario III ($n=3$ seeds).

Setting	Flow $ \Delta $		Safety $ \Delta $			
	Throughput	Halting	Collisions	Em. brakes	TTC P5 (s)	ε scale
<i>Sweep η_P (with $\kappa_E = 0.20$, $\eta_N = 0.05$ fixed)</i>						
$\eta_P = 0.05$	0.00	0.000	0	4.67	0.000	$1.0\times$
$\eta_P = 0.10$	0.00	0.013	0	4.67	0.000	$1.5\times$
$\eta_P = 0.20$	0.00	0.020	2	3.67	0.000	$2.5\times$
<i>Sweep κ_E (with $\eta_P = 0.10$, $\eta_N = 0.05$ fixed)</i>						
$\kappa_E = 0.05$	0.00	0.000	0	0.00	0.000	$0.25\times$
$\kappa_E = 0.10$	0.00	0.000	0	0.00	0.000	$0.50\times$
$\kappa_E = 0.20$	0.00	0.000	0	4.67	0.000	$1.0\times$
$\kappa_E = 0.40$	0.00	0.000	0	4.67	0.000	$2.0\times$
<i>Sweep η_P realised as SUMO LC’s lcSpeedGain on DAMA-on-SUMO ($\kappa_E = 0.20$ fixed)</i>						
lcSpeedGain = 1.5	2.33	—	0	31	—	$0.5\times$
lcSpeedGain = 2.0	0.67	—	0	10	—	$0.7\times$
lcSpeedGain = 3.0	0.67	—	0	13	—	$1.0\times$
lcSpeedGain = 5.0	0.33	—	0	17	—	$1.7\times$

Table 16: Warehouse multi-robot nominal performance ($n=5$ seeds, 20×3 grid, single chokepoint, 60 robots over 80 steps). DAMA operates within the base controller’s noise band, neither degrading nor amplifying it.

Method	Throughput $\uparrow$	Avg. wait at chokepoint $\downarrow$
Base (A^* + chokepoint arbitration)	42.60 ± 3.85	2.57
DAMA ($E_k + P_k$ via warehouse channels)	41.60 ± 3.65	2.40
DAMA – Base	-1.00 ($z = -0.42$, n.s.)	-0.17

Table 17: Warehouse hallucination robustness ($n=5$ seeds, DAMA only). Maximum $|\Delta|$ from nominal across H1–H5 is 3.0 throughput units (7% of nominal).

Hallucination mode	DAMA throughput	Δ vs nominal
Nominal (no hallucination)	41.60	ref
H1 “all clear” (under-report)	42.60	+1.00
H2 “phantom jam” (over-report)	42.40	+0.80
H3 mis-target (CA-side)	44.60	+3.00
H4 drift	42.60	+1.00
H5 stale-context	42.60	+1.00
max $ \Delta $	—	3.00 (7%)

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and Section 1 enumerate four contributions, and each of which is directly substantiated by Section 3 (architecture), Section 4 (Lemma 1 and Theorem 1), and Section 5 (six in-distribution scenarios, five adversarial and five hallucination drivers, three base controllers, two unseen geometries, and a multi-robot domain).

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Justification: Training details for the MAPPO backbone are in Appendix 8.7 (Table 13: optimizer Adam, $\text{lr } 3 \times 10^{-4}$, 5×10^5 training transitions, GAE $\lambda = 0.95$, PPO clip 0.2, network architecture, observation/critic decomposition, DAMA-aware shaping schedule, $\zeta_2 \sim U[0.15, 0.95]$). Test scenarios (Table 1) and hyperparameter sweeps validating the bound (Appendix 8.9, Table 15) are also reported.

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